PROJECT COOL POWER CORE

Automated Closed-Loop Atmospheric Remediation Engine & Multi-Frontier Micro-Purification Architecture

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1. Introduction & Physical Mechanics

Project Cool Power Core is a self-powering atmospheric remediation framework that utilizes non-thermal plasma chemistry, extreme thermodynamic boundary manipulation, and frictionless continuum fluid mechanics to cleanse ambient air arrays. Net electricity is treated exclusively as an extracted waste byproduct of the system cycle. By preheating incoming water volume to 90°C, the system effectively bypasses the conventional energy penalties associated with the latent heat of vaporization, moving the working fluid directly to its boiling threshold.

2. Fundamental Mathematical Formulations

2.1. Thermodynamic Vapor Tension & Latent Heat Bypass

The reduction in sensible heat requirements and the corresponding inflation of internal vapor pressure are modeled via the Clausius-Clapeyron relation. At the target preheat threshold, the mechanical tension required to induce cavitation collapses exponentially:

> **Equation 1: Fluidic Vapor Pressure Regulation**
>
> ```
> Pv(T) = Pref * exp[ (H_vap / R) * (1/T_ref - 1/T) ]
> ==> Pv(363.15 K) = 70.1 kPa
> ```

The cumulative heat duty (Q_dot) matching the counter-flow micro-channel heat exchanger matrix handles a mass feed rate of mw_dot = 1.5 x 10^-4 kg/s (0.54 L/h) across an explicit Logarithmic Mean Temperature Difference (LMTD):

> **Equation 2: Micro-Channel Thermal Sizing Matrix**
>
> ```
> Q_dot = mw_dot * Cp * delta_T = 43.93 Watts
> LMTD = (delta_T1 - delta_T2) / ln(delta_T1 / delta_T2) = 84.6°C
> ```

2.2. Plasma-Chemical Kinetics & Advanced Oxidation

The destruction profile of urban volatile organic compounds (VOCs) like Benzene (C6H6) under the 33.6 kV nanosecond pulsing corona field is governed by a diffusion-limited pseudo-first-order rate vector:

Equation 3: VOC Electrophilic Cleavage Stream

```
-d[C6H6]/dt = k_plasma * [OH_radical] * [C6H6]
where k_plasma = A * T^beta * exp(-E_a / (R * T))
```

2.3. Frictionless Concentric Gas-Film Equilibrium

The 5-micron aerostatic air bearing clearance maintaining the concentric isolation of the 25 mm piston is modeled via Navier-Stokes slit-flow approximations. The dynamic load capability functions proportionally to the third power of the physical gap parameter (h^3):

Equation 4: Slit-Flow Aerostatic Mass Flux

```
m_ab_dot = (pi * D * h^3 * (P_feed^2 - P_ambient^2)) / (24 * mu * R
* T * L)
```

2.4. Boundary-Layer Momentum Transfer (Tesla Condenser)

The co-rotating boundary layer shear stress profiles within the 0.25 mm channels of the centrifugal cyclonic condenser govern the optimum rotational angular velocity (omega_opt) for complete phase separation without disk flooding:

Equation 5: Tangential Shear Optimization & Angular Velocity

```
tau_z_theta(r) = (4 * mu / b) * [v_theta_bulk(r) - r * omega]
==> omega_opt = (4 * v_theta_in * r_out / 3) * [ (r_out^3 -
r_in^3) / (r_out^4 - r_in^4) ] = 11,888 RPM
```

3. Multi-Scale Device Parameterization Matrix

The structural mathematical model translates directly across geometric envelopes to adapt to specific operational frontiers:

Scale Classification	Physical Dimensions (Bore × Stroke)	Volumetric Air Flux	Net Thermal Load	Net Power Yield
Micro-Scale (Deep Space / Suits)	10 mm × 15 mm	2.5 m^3/hr	2.64 Watts	+15 Watts Surplus
Medium-Scale (Automotive Cabin)	25 mm × 40 mm	50.0 m^3/hr	43.93 Watts	+320 Watts Surplus
Macro-Scale (Industrial Scrubbing)	120 mm × 180 mm	5,200 m^3/hr	4,550 Watts	+45.0 Kilowatts Surplus

4. Field Diagnostics & Indefinite Lifecycle Maintenance

To preserve continuum alignment over the target 25,000-hour runtime, field operators must perform regular maintenance loops:

- **Aerostatic Air-Bearing Flush (5,000 Hours):** Run an ultra-dry high-pressure gaseous nitrogen sweep to clean out any particulate blockages from the 0.12 mm laser-drilled orifices.
- **Substrate Electrostatic Ground Audit (10,000 Hours):** Re-verify the continuity of the printed high-resistance graphene drain strips (100 kΩ/sq) over the Rogers RO4003C material to ensure static discharge lanes remain operational.
- **Micro-Channel Decalcification Pool (15,000 Hours):** Circulate a heated 10% citric acid bath at 50°C through the copper micro-channels for 180 seconds to strip off passivation scale and preserve thermal transfer coefficients.